

I design language platforms and runtimes. I created UniversalToolchain/Wist2, a modular .NET SDK with typed package composition, exact manifest binding, callable-first SSA, interpreter/CIL execution, and experimental PlanFuzz.

Platform architecture

Typed contracts, independent packages, capabilities and conflicts, deterministic routes

Verifiable semantics

Callable-first SSA, call effects and trust, verifier-gated AIR → SSA → AIR

1,465 / 1,465

Zero failures across nine packages, clean consumers, and interpreter/CIL parity

EXPERIENCE

- 2024 — present

UniversalToolchain / Wist2 — creator and primary developer

 - Defined package boundaries, artifact contracts, capability/conflict resolution, routing, and pass ordering.
 - Implemented exact manifest binding, lock identity, runtime-session lifecycle ownership, and fail-closed policies.
- July 1 — August 31, 2026

MCST — compiler engineering intern

 - Developing an LLVM optimization pass and testing graph algorithms.
- 2023 — 2025

Mama Lama — Software Developer

 - Independently designed the architecture and modular structure of event applications and took them from requirements to finished products.

PROJECTS

- UniversalToolchain / Wist2

1,465/1,465 tests with zero failures across nine packages and clean-consumer projects, plus interpreter/CIL parity and reproducible configuration checks.
- PlanFuzz

Reproducible artifacts and explicit separation of confirmed, unstable, inconclusive, and infrastructure outcomes.
- PS-form Memory Dependence Analyzer

Ranked 1st of 49 with 104/104 tests, with reproducible wrong-answer and time-limit counterexamples.

SKILLS

- Language architecture

typed artifacts, package graphs, capabilities/conflicts, routes, passes, manifests, and lock identity
- IR / SSA

Bytecode/AIR, callable-first SSA, CFG, dominance, folding, SCCP-lite, unreachable elimination, DCE
- Runtime

interpreter/CIL, typed delegates, lifecycle ownership, fail-closed policies, backend parity
- Verification

differential/metamorphic testing, exact oracles, replay, reducers, sanitizers, clean consumers

EDUCATION

Software Engineering student at HSE University

RECOGNITION

- The LangDev 2026 Program Committee accepted a talk on the architecture of UniversalToolchain/Wist2 for presentation in Torremolinos, Spain.

Moscow / remote